

Physics-Informed Neural Networks for Solving the DC Optimal Power Flow Problem

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Motivation

- OPF is critical for economic dispatch and grid security, but repeatedly solving OPF can be slow.
- Supervised ML can be fast but may output infeasible solutions (violating balance, thermal limits, generator bounds).
- PINNs embed KKT structure into learning to improve feasibility while retaining speed.

DC-OPF: Problem Formulation

For generator set $\mathcal{X}$, buses $\mathcal{N}$, lines $\mathcal{L}$:

$$\min_{\{p_n\}_{n \in \mathcal{X}}} \sum_{n \in \mathcal{X}} C_n(p_n) \quad (1)$$

$$\text{s.t.} \quad \sum_{n \in \mathcal{N}} (p_n - d_n) = 0, \quad (2)$$

$$\underline{p}_n \leq p_n \leq \bar{p}_n, \quad \forall n \in \mathcal{X}, \quad (3)$$

$$\underline{o}_{l_{ij}} \leq \sum_{n \in \mathcal{N}} \psi_{l_{ij}}^n (p_n - d_n) \leq \bar{o}_{l_{ij}}, \quad \forall l_{ij} \in \mathcal{L}. \quad (4)$$

- $\psi_{l_{ij}}^n$: PTDF mapping injection at bus n to flow on line l_{ij} .

KKT Conditions for DCOPF

Duals:

λ (Power balance),

$\underline{\tau}_n, \bar{\tau}_n$ (generator limit),

$\underline{\mu}_{l_{ij}}, \bar{\mu}_{l_{ij}}$ (line limit)

1) Stationarity

$$\frac{\partial C_n}{\partial p_n} + \lambda - \underline{\tau}_n + \bar{\tau}_n + \sum_{l_{ij}} (\bar{\mu}_{l_{ij}} - \underline{\mu}_{l_{ij}}) \psi_{l_{ij}}^n = 0$$

2) Primal Feasibility

$$\sum_n (p_n - d_n) = 0$$

$$\underline{p}_n \leq p_n \leq \bar{p}_n$$

$$\underline{o}_{l_{ij}} \leq \sum \psi_{l_{ij}}^n (p_n - d_n) \leq \bar{o}_{l_{ij}}$$

3) Dual Feasibility

$$\underline{\tau}_n, \bar{\tau}_n \geq 0, \quad \underline{\mu}_{l_{ij}}, \bar{\mu}_{l_{ij}} \geq 0$$

4) Complementary Slackness

$$\lambda \left(\sum_{n \in \mathcal{N}} (p_n - d_n) \right) = 0,$$

$$\underline{\tau}_n (\underline{p}_n - p_n) = 0$$

$$\bar{\tau}_n (p_n - \bar{p}_n) = 0$$

$$\bar{\mu}_{l_{ij}} \left(\sum_n \psi_{l_{ij}}^n (p_n - d_n) - \bar{o}_{l_{ij}} \right) = 0$$

$$\underline{\mu}_{l_{ij}} \left(\underline{o}_{l_{ij}} - \sum_n \psi_{l_{ij}}^n (p_n - d_n) \right) = 0$$

Network Diagram

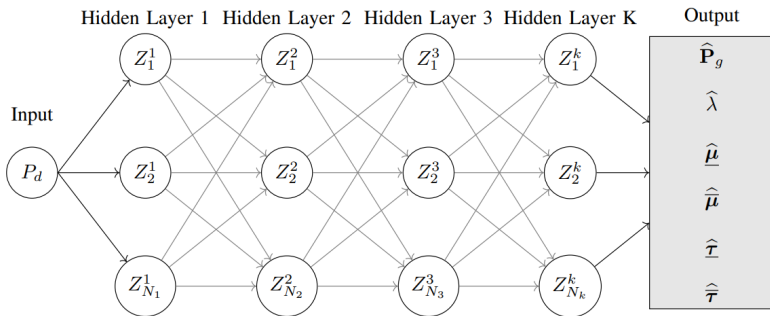


Figure: Neural network architecture

- Fully connected feed-forward network with K hidden layers.
- Layer map:

$$Z_{k+1} = \pi(W_{k+1}Z_k + b_{k+1}), \quad \hat{Z}_{k+1} = W_{k+1}Z_k + b_{k+1},$$

$$Z_{k+1} = \tanh(\hat{Z}_{k+1}).$$

From KKT Conditions to Physics-Informed Learning

Goal: Train a neural network that predicts OPF solutions while *respecting physics*.

- The KKT system encodes all optimality conditions of DC-OPF.
- Instead of enforcing KKT constraints explicitly, we convert each KKT equation into a **residual**.
- During training, the network:
 - minimizes supervised errors (matching OPF labels),
 - and minimizes KKT residuals (physics consistency).
- This yields a **PINN loss** that penalizes violations of:
 - stationarity,
 - primal feasibility,
 - dual feasibility,
 - complementary slackness.

Per-sample KKT residuals

Let $\mathcal{T}$ be the training set, $N_t = |\mathcal{T}|$. For sample $k \in \mathcal{T}$ the predictions are:

$$\hat{p}_n^{(k)}, \quad \hat{\lambda}^{(k)}, \quad \hat{\underline{\tau}}_n^{(k)}, \hat{\bar{\tau}}_n^{(k)}, \hat{\underline{\mu}}_{l_{ij}}^{(k)}, \hat{\bar{\mu}}_{l_{ij}}^{(k)}.$$

Stationarity residual:

$$r_{n,\text{stat}}^{(k)} = \frac{\partial C_n(\hat{p}_n^{(k)})}{\partial p_n} + \hat{\lambda}^{(k)} - \hat{\underline{\tau}}_n^{(k)} + \hat{\bar{\tau}}_n^{(k)} + \sum_{l_{ij} \in \mathcal{L}} (\hat{\bar{\mu}}_{l_{ij}}^{(k)} - \hat{\underline{\mu}}_{l_{ij}}^{(k)}) \psi_{l_{ij}}^n.$$

Dual feasibility residual:

$$\begin{aligned} d_{\text{df}}^{(k)} = & \sum_{n \in \mathcal{X}} \left(\max\{0, -\hat{\underline{\tau}}_n^{(k)}\} + \max\{0, -\hat{\bar{\tau}}_n^{(k)}\} \right) \\ & + \sum_{l_{ij} \in \mathcal{L}} \left(\max\{0, -\hat{\underline{\mu}}_{l_{ij}}^{(k)}\} + \max\{0, -\hat{\bar{\mu}}_{l_{ij}}^{(k)}\} \right). \end{aligned}$$

Per-sample KKT residuals

Primal feasibility residuals:

$$v_{\text{bal}}^{(k)} = \left| \sum_{n \in \mathcal{N}} (\hat{p}_n^{(k)} - d_n^{(k)}) \right|.$$

$$v_{n,\ell}^{(k)} = \max\{0, \underline{p}_n - \hat{p}_n^{(k)}\}, \quad v_{n,u}^{(k)} = \max\{0, \hat{p}_n^{(k)} - \bar{p}_n\}.$$

$$v_{l_{ij},u}^{(k)} = \max\{0, f_{l_{ij}}^{(k)} - \bar{o}_{l_{ij}}\}, \quad v_{l_{ij},\ell}^{(k)} = \max\{0, \underline{o}_{l_{ij}} - f_{l_{ij}}^{(k)}\}.$$

Complementary slackness:

$$c_{\text{bal}}^{(k)} = \hat{\lambda}^{(k)} \cdot \sum_{n \in \mathcal{N}} (\hat{p}_n^{(k)} - d_n^{(k)}).$$

$$c_{n,\ell}^{(k)} = \hat{\tau}_n^{(k)} (\underline{p}_n - \hat{p}_n^{(k)}), \quad c_{n,u}^{(k)} = \hat{\tau}_n^{(k)} (\hat{p}_n^{(k)} - \bar{p}_n),$$

$$c_{l_{ij},u}^{(k)} = \hat{\mu}_{l_{ij}}^{(k)} (f_{l_{ij}}^{(k)} - \bar{o}_{l_{ij}}), \quad c_{l_{ij},\ell}^{(k)} = \hat{\mu}_{l_{ij}}^{(k)} (\underline{o}_{l_{ij}} - f_{l_{ij}}^{(k)}).$$

MAE (averaged over training set)

All MAE terms average absolute residuals over $\mathcal{T}$:

$$\text{MAE}_p = \frac{1}{N_t} \sum_k \sum_{n \in \mathcal{X}} |\hat{p}_n^{(k)} - p_n^{(k),\text{true}}|, \quad \text{MAE}_\lambda = \frac{1}{N_t} \sum_k |\hat{\lambda}^{(k)} - \lambda^{(k),\text{true}}|,$$

$$\text{MAE}_\tau = \frac{1}{N_t} \sum_k \sum_{n \in \mathcal{X}} (|\hat{\underline{\tau}}_n^{(k)} - \underline{\tau}_n^{(k),\text{true}}| + |\hat{\bar{\tau}}_n^{(k)} - \bar{\tau}_n^{(k),\text{true}}|),$$

$$\text{MAE}_\mu = \frac{1}{N_t} \sum_k \sum_{l_{ij} \in \mathcal{L}} (|\hat{\underline{\mu}}_{l_{ij}}^{(k)} - \underline{\mu}_{l_{ij}}^{(k),\text{true}}| + |\hat{\bar{\mu}}_{l_{ij}}^{(k)} - \bar{\mu}_{l_{ij}}^{(k),\text{true}}|),$$

$$\text{MAE}_{\text{stat}} = \frac{1}{N_t} \sum_k \sum_{n \in \mathcal{X}} |r_{n,\text{stat}}^{(k)}|, \quad \text{MAE}_{\text{df}} = \frac{1}{N_t} \sum_k d_{\text{df}}^{(k)},$$

$$\text{MAE}_{\text{pf}} = \frac{1}{N_t} \sum_k \left(|v_{\text{bal}}^{(k)}| + \sum_{n \in \mathcal{X}} (v_{n,\ell}^{(k)} + v_{n,u}^{(k)}) + \sum_{l_{ij}} (v_{l_{ij},\ell}^{(k)} + v_{l_{ij},u}^{(k)}) \right),$$

$$\text{MAE}_{\text{cs}} = \frac{1}{N_t} \sum_k \left(|c_{\text{bal}}^{(k)}| + \sum_{n \in \mathcal{X}} (|c_{n,\ell}^{(k)}| + |c_{n,u}^{(k)}|) + \sum_{l_{ij}} (|c_{l_{ij},\ell}^{(k)}| + |c_{l_{ij},u}^{(k)}|) \right).$$

Total PINN Loss

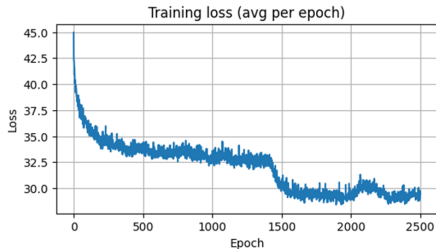
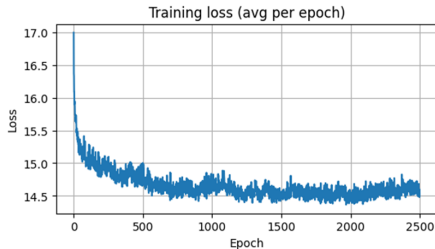
$$\mathcal{L}_{\text{PINN}} = \text{MAE}_p + \text{MAE}_\lambda + \text{MAE}_\tau + \text{MAE}_\mu \\ + \alpha_{\text{stat}}\text{MAE}_{\text{stat}} + \alpha_{\text{pf}}\text{MAE}_{\text{pf}} + \alpha_{\text{df}}\text{MAE}_{\text{df}} + \alpha_{\text{cs}}\text{MAE}_{\text{cs}}.$$

- Scalars $\alpha_i > 0$ balance supervised vs physics terms.
- Two experimental configurations: Case 1 (all $\alpha = 0$) and Case 2 (nonzero physics weights).

Dataset and Training

- IEEE 39-bus New England system. PTDF from MATPOWER. OPF solved with YALMIP + GUROBI.
- Loads sampled by Latin Hypercube Sampling ($\pm 20\%$); feasible OPF cases retained.
- Network: 3 hidden layers, 128 neurons each, Tanh activations. Output layer uses ReLU for inequality duals.
- Training: Adam, $LR = 10^{-2}$, batch size 64, 2500 epochs. Best validation model chosen.
- Test set size: $N_{\text{test}} = 1996$.

Training Loss (Case 1 vs Case 2)



Left: Case 1 (no physics). Right: Case 2 (physics-informed).

Case 2 training loss is larger (more residuals to minimize); suggests longer training, adaptive weighting, or stronger optimizers may improve results further

Constraint Violations: Case 1 (no physics)

Violation Type	Count	Total MW	Avg MW / sample
Generator capacity violations	24	564.81	0.28
Line flow violations	1750	50,515.97	25.31
Power balance violations	1956	86,928.65	45.35

Constraint Violations: Case 2 (physics-informed)

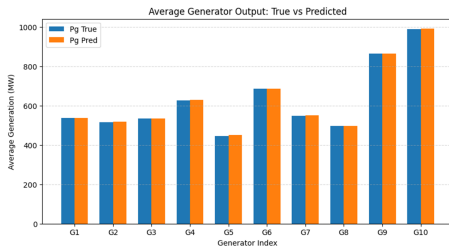
Violation Type	Count	Total MW	Avg MW / sample
Generator capacity violations	6	97.40	0.05
Line flow violations	729	11,915.03	5.97
Power balance violations	1909	37,819.57	18.95

Reduction of Violation from Case 1 to Case 2

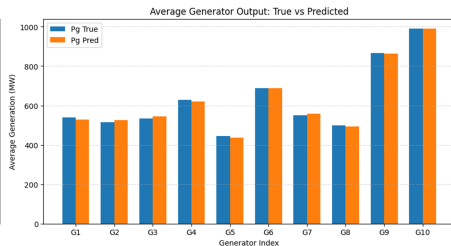
Violation Type	Count (%)	Total MW (%)	Avg MW/sample (%)
Generator Capacity	75.00%	82.76%	82.14%
Line Flow	58.34%	76.41%	76.41%
Power Balance	2.40%	56.49%	58.21%

- PINN (Case 2) drastically reduces violation *magnitudes*: generator-capacity MWs down by $\sim 82.8\%$, line-flow MWs by $\sim 76.4\%$.
- Counts of power-balance violations change little, but their magnitudes fall substantially — PINN shrinks severity of mismatches.

Generator Dispatch Comparison



Average dispatch, Case 1.



Average dispatch, Case 2.

- Adding KKT-informed physics to the loss significantly improves feasibility.
- Supervised prediction error may increase slightly, but the trade-off is worthwhile.
- In real power-system operations, feasibility is essential for safety and reliability.
- Physics-informed training helps the model respect operational constraints.

Conclusions & Future Work

- Embedding KKT residuals into the learning model effectively combines domain knowledge with data.
- PINN results show large reductions in constraint violations while maintaining strong dispatch accuracy.
- Important future directions:
 - Hyperparameter tuning and adaptive loss weighting,
 - Longer or more advanced training strategies,
 - Robustness evaluation under unseen or stressed conditions.
- PINN predictions can potentially be used to warm-start OPF solvers for faster computation.

Thank you!

I am happy to take any questions.

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